

COSC 594/650

Optimality

Dongsheng Ding

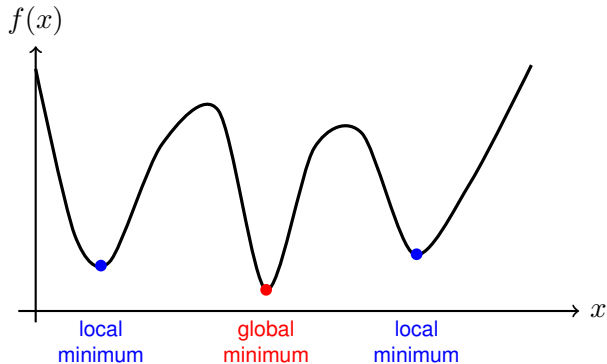
dongshed@utk.edu

Department of Electrical Engineering and Computer Science
University of Tennessee, Knoxville

Optimization

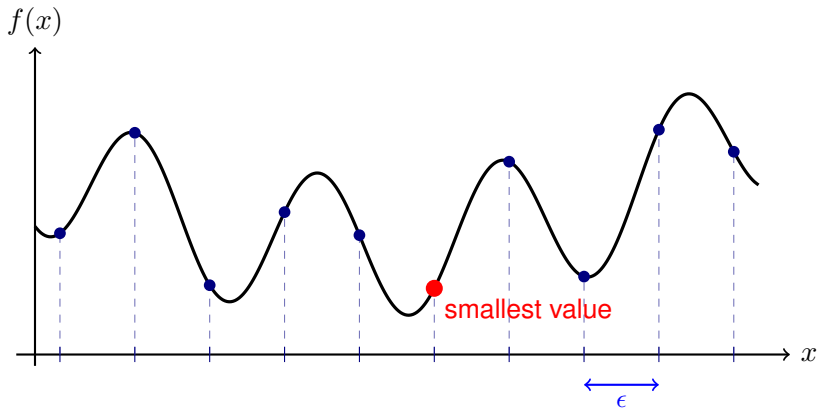
$$\begin{array}{ll} \underset{x \in \mathbb{R}^n}{\text{minimize}} & f(x) \\ \text{subject to} & x \in \mathbb{R}^n \end{array}$$

GOAL: Find a **best** decision



Grid search in one dimension

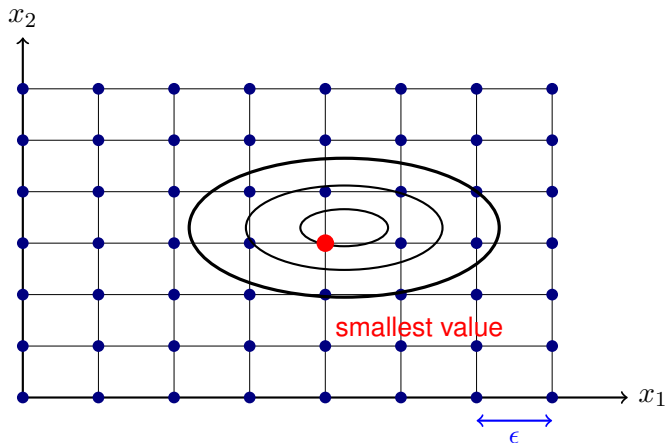
Evaluate every grid point of $x \in [-100, 100]$



Complexity: $\mathcal{O}(1/\epsilon)$ evaluations

Grid search in higher dimensions

Evaluate every grid point of $x \in [-100, 100] \times [-100, 100]$



Complexity: $\mathcal{O}(1/\epsilon^2)$ evaluations

Exponential complexity $\mathcal{O}(1/\epsilon^n)$ for n dimensions



Unconstrained optimization

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad f(x)$$

■ GLOBAL MINIMUM

$$f(x) \geq f(x^*), \quad \forall x \in \mathbb{R}^n$$

■ STRICT GLOBAL MINIMUM

$$f(x) > f(x^*), \quad \forall x \neq x^*$$

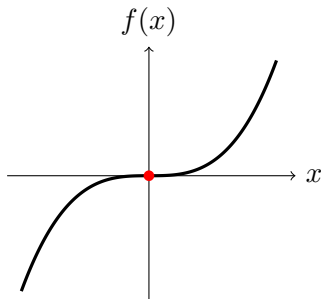
■ LOCAL AND STRICT LOCAL MINIMUM

$$\exists \epsilon > 0, \quad f(x) \geq f(x^*), \quad \forall x \in \{x \in \mathbb{R}^n \mid \|x - x^*\|_2 \leq \epsilon\}$$

with $>$ for every nearby $x \neq x^*$ in the strict case

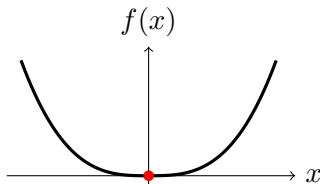
Geometrical interpretation

$$f(x) = x^3$$



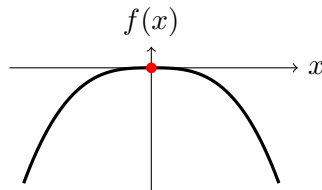
Not a minimum

$$f(x) = |x|^3$$



Strict global minimum

$$f(x) = -|x|^3$$



Strict global maximum

QUESTION: How to determine the optimality of a given point?

Local optimality conditions

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (twice continuously differentiable)

■ NECESSARY OPTIMALITY CONDITION

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succeq 0$$

★ $f(x) = x^3$, $f'(0) = f''(0) = 0$, but 0 is not a local minimum

■ SUFFICIENT OPTIMALITY CONDITION

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succ 0$$

PROOF: Taylor expansion

Why do optimality conditions matter?

■ TRACTABLE TEST

- ★ check candidate solutions
- ★ eliminate points that cannot be optimal

■ ALGORITHMIC ROLE

- ★ define stopping criteria
- ★ guide algorithm design and analysis
- ★ provide optimality certificates

Existence of an optimal solution

Function: $f: \mathbb{R}^n \rightarrow \mathbb{R}$ (continuous)

Domain: $X \subseteq \mathbb{R}^n$

■ WEIERSTRASS THEOREM

Every continuous function f attains its infimum over a compact set $X \subseteq \mathbb{R}^n$:

$$\exists x^* \in X, \quad f(x^*) = \inf_{x \in X} f(x)$$

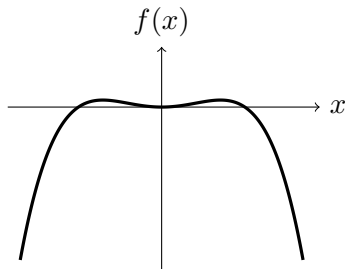
★ compact nonempty set $\{x \in \mathbb{R}^n \mid f(x) \leq f(x_0)\}$ – sufficient condition

■ COERCIVITY SUFFICIENT CONDITION

$$f(x) \rightarrow +\infty \quad \text{as} \quad \|x\|_2 \rightarrow \infty$$

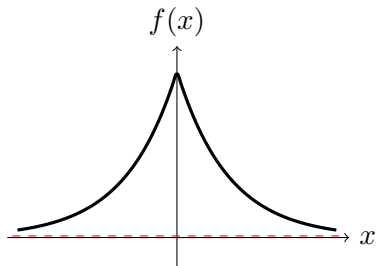
Nonexistence examples

$$f(x) = x^2 - x^4$$



No global minimum

$$f(x) = e^{-|x|}$$



No minimum is attained

Unconstrained quadratic optimization

$$\underset{x \in \mathbb{R}^n}{\text{minimize}} \quad \frac{1}{2}x^\top Qx + b^\top x + c, \quad Q = Q^\top$$

■ NECESSARY CONDITION

$$\nabla f(x^*) = 0 \quad \text{and} \quad \nabla^2 f(x^*) \succeq 0$$

- ★ $Q \succeq 0$ and $b \in \mathcal{R}(Q)$ – every solution of $Qx = -b$ is globally optimal
- ★ $Q \succeq 0$ and $b \notin \mathcal{R}(Q)$ – the problem is unbounded below

■ SUFFICIENT CONDITION

$$\nabla f(x) = 0 \quad \text{and} \quad \nabla^2 f(x) \succ 0$$

- ★ $Q \succ 0$ – unique global minimizer $x^* = -Q^{-1}b$

■ NONEXISTENCE $Q \not\succeq 0$

Questions?

(**Email: dongshed@utk.edu**)